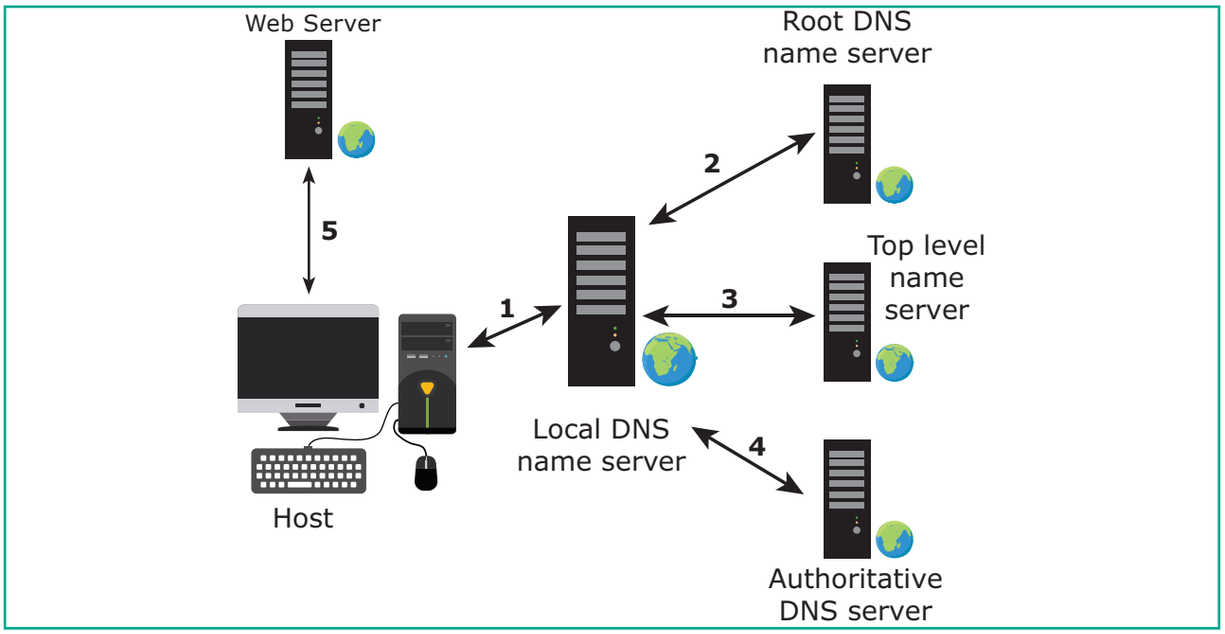


Figure 12.8 working structure of Name server

Types of Name Servers

There are three types of Name Servers which control the entire Domain Name System:

1. Root Name Server - top level server which contains entire DNS tree, maintained by ICANN. There are 13 servers.
2. Primary/Master Name Server– contains a zone resource records. These records are updatable by domain name holders such as organizations.
3. Secondary/Slave Name Server – contains a copy of primary server files. This server has no authority to update, but reduce the workload of master server by sharing the queries.

DO YOU KNOW?

ICANN, Internet Corporation for Assigned Name and Numbers is the Non-profit Organization which assigns names and numbers for all Internet resources like domain names and IP addresses.

12.5.3 Zone

The entire name space is divided into many different zones. It is the area up to which the server has access. Zone is defined as a group of contiguous domains and sub domains. If the zone has a single domain, then zone and domain are the same.

Every zone has the server which contains a database called zone file. Using the zone file, the DNS server replies to the queries about hosts in its zone. There are two copies of zone files available, Master file and slave file. Refer Figure 12.9

LET'S RECAP

- A domain is a single node of the Domain Namespace.
- A zone is a subset of the Domain namespace generally stored in a file.
- Domain Name space is an entire collection Domains, Sub domains and Zones
- Name server manages the database of domain names and corresponding IP addresses.
- A server can contain more than one zone files (Zones). A zone can contain more than one sub domains

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